

Selective Mg Insertion into Substituted Mono- and Dichloro Arenes in the Presence of LiCl: A New Preparation of Boscalid

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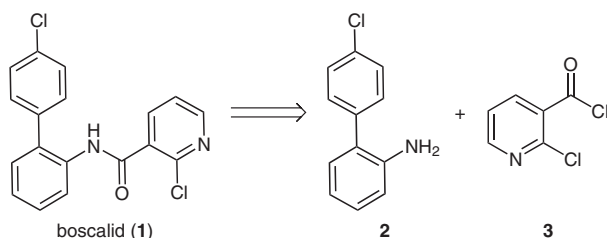
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Dedicated to Prof. Alfredo Ricci on the occasion of his retirement

Abstract: The LiCl-mediated Mg insertion into polysubstituted aryl chlorides bearing up to three chloro substituents in *ortho* or *meta* position selectively leads to Grignard reagents which readily react with various electrophiles. As an application, we have developed a new formal synthesis of boscalid.

Key words: insertion, magnesium, lithium, aryl chlorides, boscalid

The use of substituted aryl chlorides as starting materials for the preparation of organometallic intermediates is of interest since they are inexpensive and usually commercially available (compared to the corresponding aryl iodides or bromides). Of special interest is the preparation of Grignard reagents derived from substituted di- or trichloroarenes. Such organometallics may react further with various electrophiles and provide aromatic derivatives of biological interest such as boscalid (**1**), which controls different plant pathogens in horticultural crops.^{1a} A retrosynthesis of boscalid (**1**) requires an efficient preparation of the aniline **2**. The nicotinyl chloride **3** is readily available (Scheme 1).¹ The biphenyl **2** may be readily prepared using a 4-chloro-substituted Grignard reagent.

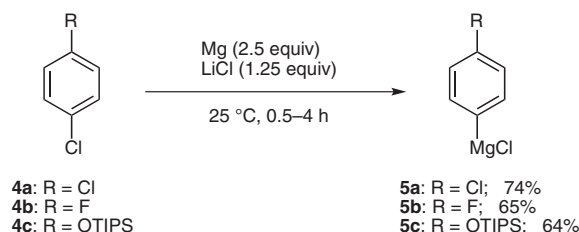


Scheme 1 Retrosynthesis of boscalid (**1**)

Several procedures starting from chloro-substituted aromatics affording the corresponding Grignard reagents have already been described.^{2–4} The use of Rieke-Mg which is prepared from MgCl_2 and lithium naphthalenide (20 mol%) results in insertion reaction with various aryl halides under especially mild conditions.⁵

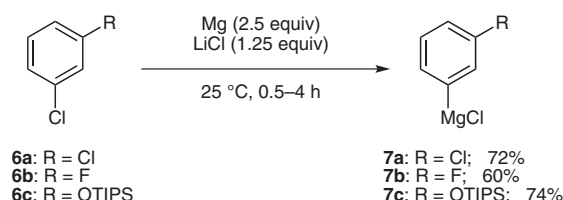
Recently, we have described a new LiCl-mediated magnesium insertion into aryl and heteroaryl chlorides.⁶ The presence of LiCl considerably facilitates the magnesium

insertion and allows the use of low reaction temperatures in large-scale preparations of organomagnesium reagents. Furthermore, LiCl is an inexpensive salt which may often also be used in catalytic amounts (30–50 mol%). Herein, we wish to report a convenient magnesium insertion procedure in the presence of LiCl into various mono- and polychlorinated arenes. Thus, the reaction of various 4-substituted chloroarenes of type **4** with Mg (2.5 equiv) and LiCl (1.25 equiv) was complete within a few hours at 25 °C (Scheme 2). Control experiments performed without LiCl showed much lower rates for the magnesium insertion. Interestingly, 1,4-dichlorobenzene (**4a**) afforded selectively the mono-insertion magnesium reagent **5a** in 74% yield.⁷ Also, the aryl chlorides **4b,c** bearing an F or an OTIPS substituent in *para* position underwent clean insertion reactions leading to the desired organomagnesium compounds **5b,c** in 64–65% yield.



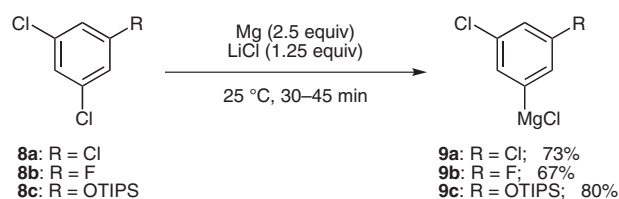
Scheme 2 Mg insertion into *para*-substituted aryl chlorides **4a–c**

Analogously, 3-substituted aryl chlorides of type **6** reacted with Mg turnings and LiCl to give the Grignard reagents **7a–c** (Scheme 3). The reaction conversions were complete within 0.5–4 hours. Therefore, (3-chlorophenyl)magnesium chloride (**7a**), (3-fluorophenyl)magnesium chloride (**7b**) and {3-[(triisopropylsilyl)oxy]phenyl}magnesium chloride (**7c**) were prepared from the *meta*-substituted aryl chlorides **6a–c** in 60–74% yield.



Scheme 3 Mg insertion into *meta*-substituted aryl chlorides **6a–c**

Additionally, the magnesium insertion to substituted 1,3-dichlorobenzenes of type **8** was performed in less than one hour (Scheme 4). The magnesium reagents substituted with a chlorine, fluorine or (triisopropylsilyl)oxy group **9a–c** were obtained in 67–80% yield. Double magnesium insertion in 1,3,5-trichlorobenzene was not observed. The freshly prepared magnesium reagents⁸ reacted with the usual electrophiles (Table 1).

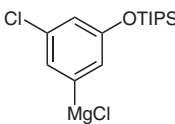
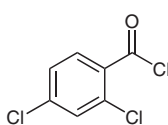
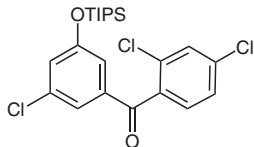


Scheme 4 Mg insertion in *meta*-substituted 1,3-dichloroarenes

Table 1 Trapping of the Mg Reagents with Various Electrophiles

Entry	Grignard reagent (h) ^a	Electrophile	Product, Yield ^b
1	 5b (2)		 10: 73% ^c
2	 5c (4)		 11: 81%
3	 7a (0.75)		 12: 85% ^{d,e}
4	 7b (4)		 13: 74% ^{d,e}
5	 7c (3)		 14: 96%
6	 9a (0.5)	CO ₂	 15: 99%
7	 9b (0.75)		 16: 91% ^{d,e}

Table 1 Trapping of the Mg Reagents with Various Electrophiles (continued)

Entry	Grignard reagent (h) ^a	Electrophile	Product, Yield ^b
8	 9c (0.75)		 17 : 80% ^c

^a Reaction time at 25 °C.^b Isolated yield of analytically pure product.^c CuCN·2LiCl (1.1 equiv) was used.^d Yield after transmetalation with ZnCl₂.^e Pd(dba)₂ (3%) and P(*o*-furyl)₃ (6%) were used.

Thus, acylation of **5b** in the presence of CuCN·2LiCl (1.1 equiv)⁹ with thiophene-2-carbonyl chloride gave the ketone **10**¹⁰ in 73% yield (entry 1). The reaction of {4-[(triisopropylsilyloxy)phenyl]magnesium chloride (**5c**) with dimethylcarbamoyl chloride yielded the amide **11**¹¹ in 81% yield (entry 2). Pd-catalyzed cross-coupling reaction of (3-chlorophenyl)magnesium chloride (**7a**) or (3-fluorophenyl)magnesium chloride (**7b**) with heteroaryl iodides afforded the heterocycles **12**¹² and **13**¹³ in 74% and 85% yield, respectively (entries 3 and 4). Formylation using DMF or addition of CO₂¹⁴ to **7c** and **9a** provided 3-[(triisopropylsilyloxy)benzaldehyde (**14**)¹⁵ and 3,5-dichlorobenzoic acid (**15**)¹⁶ in 96% and 99% yield, respectively (entries 5 and 6). The Pd-catalyzed cross-coupling reaction of 3-iodobenzonitrile with **9b** yielded the biaryl **16**¹⁷ in 91% yield (entry 7). Copper-catalyzed acylation of **9c** with 2,4-dichlorobenzoyl chloride led to the ketone **17**¹⁸ in 80% yield (entry 8). Negishi cross-coupling^{19,20} of (4-chlorophenyl)zinc chloride (obtained after transmetalation of **5a** with ZnCl₂) with 2-bromoaniline afforded 4'-chloro-(1,1'-biphenyl)-2-amine (**2**)²¹ in 70% yield, which can be used as a precursor for the synthesis of boscalid (**1**) (Scheme 5).¹

The use of SPhos²² as catalyst makes the Pd-catalyzed cross-coupling very fast and avoids therefore the need to protect the amino function.²⁰ We could, therefore, prepare boscalid (**1**) in two steps from the readily available starting materials [1,4-dichlorobenzene (**4a**), 2-bromoaniline, 2-chloronicotinoyl chloride (**3**), overall yield: 66%].

In conclusion, we have developed an efficient and practical insertion procedure in the presence of LiCl using commercial magnesium turnings into substituted aryl chlorides leading to the corresponding Grignard reagents

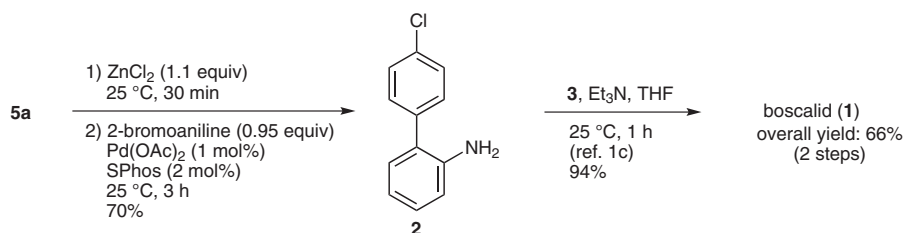
in 60–80% yield. These reactions can be readily scaled up.^{6b} We could thus show various trapping reactions with different electrophiles leading to products which can be applied for instance to the synthesis of boscalid. Furthermore, this method gives an environmentally friendly access to simple substituted aryl scaffolds starting from inexpensive aryl chlorides, which is of great use for the chemical and pharmaceutical industry.

Acknowledgment

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References and Notes

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**Scheme 5** Synthesis of 4'-chloro-(1,1'-biphenyl)-2-amine (**2**)

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- (7) The yields of the Grignard reagents were determined by titration using I_2 ; Krasovskiy, A.; Knochel, P. *Synthesis* **2006**, 890.
- (8) **Preparation of the Grignard Reagents (TP1)**: A dry and argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with LiCl (530 mg, 12.5 mmol) and heated with a heat gun under high vacuum for 10 min. Magnesium turnings (608 mg, 25 mmol) and THF (10 mL) were added and the magnesium was activated with 1,2-dibromoethane (1 drop per mmol). After 5 min of stirring, the aryl chloride (10 mmol) was added and the reaction mixture was stirred for the indicated time at 25 °C. GC analysis of a quenched reaction aliquot showed complete conversion. Then, the supernatant solution was cannulated to a new, dry and argon-flushed Schlenk flask and the yield was determined by iodometric titration.
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- (10) **Preparation of (4-Fluorophenyl)(thiophen-2-yl)-methanone (10)**: The freshly prepared and titrated (4-fluorophenyl)magnesium chloride (**4b**; 1.59 mL, 1 mmol) prepared according to TP1 (2 h, 0.63 M, 65%), was transmetalated with CuCN·2LiCl (1 M in THF, 1.1 mL, 1.1 mmol) at –20 °C and the mixture was stirred for 30 min. After addition of thiophene-2-carbonyl chloride (117 mg, 0.8 mmol) the resulting solution was stirred for 30 min at this temperature. The reaction was then quenched with sat. aq NH_4Cl solution (10 mL) and the resulting mixture was extracted with Et_2O (3 × 10 mL). The combined organic layers were dried over $MgSO_4$ and concentrated in vacuo. The crude residue was purified by flash column chromatography (iso-hexane– Et_2O , 50:1) to give **10** as a white solid (151 mg, 73%). (b) Analytical data matches the literature: Zhao, W.; Carreira, E. M. *Chem. Eur. J.* **2007**, 13, 2671.
- (11) **Preparation of *N,N*-Dimethyl-4-[(triisopropylsilyl)oxy]benzamide (11)**: To the freshly prepared and titrated {4-[(triisopropylsilyl)oxy]phenyl}magnesium chloride (**4c**; 1.56 mL, 1 mmol) prepared according to TP 1 (4 h, 0.64 M, 64%), was added dimethylcarbamoyl chloride (86 mg, 0.8 mmol) at –20 °C and the mixture was stirred for 30 min. The reaction was then quenched with sat. aq NH_4Cl solution (10 mL) and the resulting mixture was extracted with Et_2O (3 × 10 mL). The combined organic layers were dried over $MgSO_4$ and concentrated in vacuo. The crude residue was purified by flash column chromatography (iso-hexane– Et_2O , 10:1) to give **11** as a colorless oil (260 mg, 81%). IR (ATR): 2944, 2867, 1634, 1604, 1514, 1490, 1463, 1387, 1261, 1168, 1102, 1078, 1013, 997, 908, 882, 845, 766, 752, 709, 684, 675 cm^{-1} . 1H NMR (400 MHz, $CDCl_3$): δ = 7.29–7.33 (m, 2 H), 6.84–6.88 (m, 2 H), 3.03 (s, 6 H), 1.20–1.31 (m, 3 H), 1.08 (d, J = 7.2 Hz, 18 H). ^{13}C NMR (100 MHz, $CDCl_3$): δ = 171.6, 157.3, 129.0, 128.7, 119.5, 36.4 (br), 17.8, 12.6. MS (70 eV, EI): m/z (%) = 321 (22) [M^+], 279 (18), 278 (100), 277 (12), 250 (39), 223 (11), 222 (57), 208 (18), 111 (11), 110 (25), 103 (19), 72 (88). HRMS (EI): m/z calcd for $C_{18}H_{31}NO_2Si$: 321.2124; found: 321.2118.
- (12) **Preparation of 2-(3-Chlorophenyl)thiophene (12)**: The freshly prepared and titrated (3-chlorophenyl)magnesium chloride (**6a**; 1.33 mL, 1 mmol) prepared according to TP1 (45 min, 0.75 M, 72%), was transmetalated with $ZnCl_2$ (1 M in THF, 1.1 mL, 1.1 mmol) at 25 °C. After addition of 2-iodothiophene (168 mg, 0.8 mmol), $Pd(dba)_2$ (17 mg, 3 mol%) and $P(o-furyl)_3$ (14 mg, 6 mol%) the resulting solution was stirred for 1 h at 25 °C. The reaction mixture was then quenched with sat. aq NH_4Cl solution (10 mL) and the resulting mixture was extracted with Et_2O (3 × 10 mL). The combined organic layers were dried over $MgSO_4$ and concentrated in vacuo. The crude residue was purified by flash column chromatography (iso-hexane) to give **12** as a colorless solid (165 mg, 85%); mp 25 °C. IR (ATR): 3104, 3066, 3035, 1932, 1856, 1794, 1658, 1589, 1563, 1529, 1479, 1466, 1419, 1256, 1215, 1172, 1078, 1056, 979, 874, 856, 826, 774, 763, 752, 734, 693, 680 cm^{-1} . 1H NMR (300 MHz, $CDCl_3$): δ = 7.57–7.58 (m, 1 H), 7.44–7.48 (m, 1 H), 7.28–7.30 (m, 3 H), 7.20–7.24 (m, 1 H), 7.06 (dd, J = 5.0, 3.6 Hz, 1 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.7, 136.1, 134.8, 130.1, 128.1, 127.3, 125.9, 125.5, 124.0, 123.8. MS (70 eV, EI): m/z (%) = 196 (36), 195 (12), 194 (100) [M^+], 159 (13), 149 (16), 115 (32). HRMS (EI): m/z calcd for $C_{10}H_7ClS$: 193.9957; found: 193.9952.
- (13) **Preparation of 4-(3-Fluorophenyl)-2,6-dimethoxy-pyrimidine (13)**: The freshly prepared and titrated (3-fluorophenyl)magnesium chloride (**6b**; 1.56 mL, 1 mmol) prepared according to TP1 (4 h, 0.64 M, 60%), was transmetalated with $ZnCl_2$ (1 M in THF, 1.1 mL, 1.1 mmol) at 25 °C. After addition of 4-iodo-2,6-dimethoxypyrimidine (213 mg, 0.8 mmol), $Pd(dba)_2$ (17 mg, 3 mol%) and $P(o-furyl)_3$ (14 mg, 6 mol%) the resulting solution was stirred for 3 h at 25 °C. The reaction mixture was then quenched with sat. aq NH_4Cl solution (10 mL) and the resulting mixture was extracted with Et_2O (3 × 10 mL). The combined organic layers were dried over $MgSO_4$ and concentrated in vacuo. The crude residue was purified by flash column chromatography (iso-hexane– Et_2O , 10:1) to give **13** as a light yellow solid (173 mg, 74%); mp 53.1–54.4 °C. IR (ATR): 3080, 2954, 2361, 1567, 1481, 1462, 1352, 1258, 1242, 1199, 1160, 1099, 1018, 982, 922, 886, 825, 802, 784, 777, 706, 662 cm^{-1} . 1H NMR (300 MHz, $CDCl_3$): δ = 7.75–7.79 (m, 2 H), 7.37–7.44 (m, 1 H), 7.11–7.17 (m, 1 H), 6.74 (s, 1 H), 4.07 (s, 3 H), 4.00 (s, 3 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 172.7, 165.5, 164.5 (d, J = 2.8 Hz), 163.1 (d, J = 246 Hz), 139.1 (d, J = 7.8 Hz), 130.1 (d, J = 8.1 Hz), 122.5 (d, J = 2.8 Hz), 117.4 (d, J = 21.6 Hz), 114.0 (d, J = 23.0 Hz), 97.4, 54.8, 53.9. MS (70 eV, EI): m/z (%) = 235 (18), 234 (100) [M^+], 233 (74), 219 (18), 205 (23), 204 (56), 203 (13), 189 (24), 163 (16), 146 (15). HRMS (EI): m/z calcd for $C_{12}H_{11}FN_2O_2$: 234.0805; found: 234.0808.
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- (15) (a) **Preparation of 3-[(Triisopropylsilyl)oxy]benzaldehyde (14)**: To the freshly prepared and titrated {3-[(triisopropylsilyl)oxy]phenyl}magnesium chloride (**6c**; 1.45 mL, 1 mmol) prepared according to TP1 (3 h, 0.69 M, 74%), was added *N,N*-dimethylformamide (80 mg, 1.1 mmol) at –20 °C and the mixture was stirred for 30 min. The reaction was then quenched with sat. aq NH_4Cl solution (10 mL) and the resulting mixture was extracted with Et_2O (3 × 10 mL). The combined organic layers were dried over $MgSO_4$ and concentrated in vacuo. The crude residue was purified by flash column chromatography (iso-hexane– Et_2O , 20:1) to give **14** as a yellow oil (267 mg, 96%). (b) Analytical data matches the literature: Joncour, A.; Liu, J. M.; Décor, A.; Thoret, S.; Wdzieczak-Bakala, J.; Bignon, J.; Baudoin, O. *ChemMedChem* **2008**, 3, 1731.
- (16) (a) **Preparation of 3,5-Dichlorobenzoic Acid (15)**: The freshly prepared and titrated (3,5-dichlorophenyl)-

magnesium chloride (**8a**; 1.54 mL, 1 mmol) prepared according to TP1 (30 min, 0.65 M, 73%), was added to a flask filled with anhyd CO₂ (g). Then, anhyd CO₂ (g) was bubbled through the reaction mixture (ca. 5 min) until a balloon attached to the reaction flask by a short length rubber tubing and a needle adapter was inflated. The reaction mixture was stirred for 1 h at 25 °C and then diluted with Et₂O (15 mL) and extracted with sat. aq NaHCO₃ (3 × 20 mL). The combined aq phases were carefully acidified with HCl (5 mL) until pH < 5 and extracted with Et₂O (3 × 30 mL). The combined organic layers were dried over MgSO₄ and concentrated in vacuo to give **15** as a yellow solid (189 mg, 99%). (b) Analytical data matches the literature: Heiss, C.; Marzi, E.; Schlosser, M. *Eur. J. Org. Chem.* **2003**, 4625.

- (17) **Preparation of 3'-Chloro-5'-fluoro-(1,1'-biphenyl)-3-carbonitrile (16)**: The freshly prepared and titrated (3-chloro-5-fluorophenyl)magnesium chloride (**8b**; 1.49 mL, 1 mmol) prepared according to TP1 (45 min, 0.67 M, 67%), was transmetalated with ZnCl₂ (1 M in THF, 1.1 mL, 1.1 mmol) at 25 °C. After addition of 3-iodobenzonitrile (183 mg, 0.8 mmol), Pd(dba)₂ (17 mg, 3 mol%) and P(*o*-furyl)₃ (14 mg, 6 mol%) the resulting solution was stirred for 2 h at 25 °C. The reaction mixture was then quenched with sat. aq NH₄Cl solution (10 mL) and the resulting mixture was extracted with Et₂O (3 × 10 mL). The combined organic layers were dried over MgSO₄ and concentrated in vacuo. The crude residue was purified by flash column chromatography (isohexane–Et₂O, 10:1) to give **16** as a colorless solid (211 mg, 91%); mp 118.7–120.2 °C. IR (ATR): 307, 2362, 2338, 2231, 1588, 1574, 1444, 1396, 1324, 1308, 1276, 1202, 1175, 1164, 10094, 1069, 945, 916, 890, 866, 837, 791, 748, 694, 680 cm⁻¹. ¹H NMR (300 MHz, CDCl₃): δ = 7.74–7.81 (m, 2 H), 7.66–7.70 (m, 1 H), 7.54–7.60 (m, 1 H), 7.33–7.34 (m, 1 H), 7.11–7.17 (m, 2 H). ¹³C NMR (75 MHz, CDCl₃): δ = 162.9 (d, *J* = 250 Hz), 141.9 (d, *J* = 8.7 Hz), 139.9, 135.8 (d, *J* = 11.0 Hz), 131.8, 131.3, 130.5, 129.9, 123.2, 118.2, 116.0 (d, *J* = 25.0 Hz), 113.4, 112.6 (d, *J* = 23.0 Hz). MS (70 eV, EI): *m/z* (%) = 233 (32), 232 (14), 231 (100) [M⁺], 196 (17), 195 (22), 169 (13), 58 (11), 43 (28). HRMS (EI): *m/z* calcd for C₁₃H₇ClFN: 231.0251; found: 231.0245.

- (18) **Preparation of {3-chloro-5-[(triisopropylsilyl)-oxy]phenyl}(2,4-dichlorophenyl)methanone (17)**: The freshly prepared and titrated {3-chloro-5-[(triisopropylsilyl)-oxy]phenyl)magnesium chloride (**8c**; 1.43 mL, 1 mmol) prepared according to TP1 (45 min, 0.70 M, 80%), was transmetalated with CuCN·2LiCl (1 M in THF, 1.1 mL, 1.1 mmol) at –20 °C and the mixture was stirred for 30 min.

After addition of 2,4-dichlorobenzoyl chloride (168 mg, 0.8 mmol) the resulting solution was stirred for 30 min at this temperature. The reaction mixture was then quenched with sat. aq NH₄Cl solution (10 mL) and the resulting mixture was extracted with Et₂O (3 × 10 mL). The combined organic layers were dried over MgSO₄ and concentrated in vacuo. The crude residue was purified by flash column chromatography (isohexane–Et₂O, 50:1) to give **17** as a colorless oil (366 mg, 80%). IR (ATR): 2946, 2867, 1718, 1606, 1595, 1566, 1501, 1453, 1433, 1398, 1318, 1254, 1199, 1137, 1112, 1065, 1025, 999, 964, 921, 880, 854, 799, 762, 738, 688 cm⁻¹. ¹H NMR (300 MHz, CDCl₃): δ = 7.48 (d, *J* = 1.7 Hz, 1 H), 7.35–7.38 (m, 1 H), 7.27–7.32 (m, 2 H), 7.09–7.14 (m, 2 H), 1.16–1.28 (m, 3 H), 1.07 (d, *J* = 7.2 Hz, 18 H). ¹³C NMR (75 MHz, CDCl₃): δ = 192.8, 157.2, 138.5, 137.0, 136.3, 135.3, 132.5, 130.1, 130.1, 127.2, 125.5, 122.6, 119.2, 17.8, 12.5. MS (70 eV, EI): *m/z* (%) = 458 (8) [M⁺], 417 (43), 416 (23), 415 (98), 414 (43), 413 (100), 389 (17), 388 (11), 387 (53), 386 (16), 385 (51), 357 (16), 356 (38), 354 (41), 279 (13), 251 (25), 249 (37), 179 (15), 178 (17), 177 (11), 174 (58), 173 (84), 145 (16), 95 (44), 94 (10), 79 (14), 75 (10), 44 (15), 43 (11), 41 (17). HRMS (EI): *m/z* calcd for C₂₂H₂₇Cl₃O₂Si: 456.0846; found: 458.0842.

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- (21) **Preparation of 4'-Chloro-(1,1'-biphenyl)-2-amine (2)**:^{1b} The freshly prepared and titrated (4-chlorophenyl)-magnesium chloride (**4a**; 25 mL, 19.3 mmol) prepared according to TP1 (30 min, 0.77 M, 70%), was transmetalated with ZnCl₂ (1 M in THF, 19.3 mL, 19.3 mmol) and then slowly added to a 1 M premixed solution of 2-bromoaniline (3.15 g, 18.3 mmol), Pd(OAc)₂ (45 mg, 1 mol%), SPhos (164 mg, 1 mol%) in THF (19 mL). The resulting solution was stirred for 3 h at 25 °C. The reaction mixture was then quenched with sat. aq NH₄Cl solution (150 mL) and the resulting mixture was extracted with Et₂O (3 × 150 mL). The combined organic layers were dried over MgSO₄ and concentrated in vacuo. The crude residue was purified by flash column chromatography (CH₂Cl₂) to give **2** as a yellow solid (2.75 g, 70%).
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